Solubility Equilibria

Ksp, molar solubility and precipitation

Name _____ Class _____ Date _____

KSP VALUES AT 25 °C

$AgCl$ 1.8×10^{-10} · $BaSO_4$ 1.1×10^{-10} · CaF_2 3.9×10^{-11} · $Mg(OH)_2$ 1.8×10^{-11} · Ag_2CrO_4 1.1×10^{-12}

Molar mass $BaSO_4$ 233.39 g/mol · $K_w = 1.0 \times 10^{-14}$ · pH + pOH = 14.00

1 : 1 salts: $K_{sp} = s^2$ · 1 : 2 and 2 : 1 salts: $K_{sp} = 4s^3$

Write the dissolving equilibrium first. Allow for dilution when solutions are mixed.

A. Expressions

1. Write the dissolving equilibrium and the Ksp expression for: (a) $BaSO_4$ (b) $Mg(OH)_2$ (c) Ag_2CrO_4 (d) $Ca_3(PO_4)_2$

B. Molar solubility

2. Calculate the molar solubility of $BaSO_4$ in mol/L and in g/L.

3. Calculate (a) the molar solubility of $Mg(OH)_2$ (b) $[OH^-]$ in its saturated solution (c) the pH of the saturated solution.

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.

4. The molar solubility of Ag_2CrO_4 is 6.5×10^{-5} mol/L. Calculate its Ksp.

5. AgCl has a larger Ksp than Ag_2CrO_4. Show that Ag_2CrO_4 is nevertheless the more soluble salt, and explain why.

C. Common ion and precipitation

6. Calculate the molar solubility of CaF_2 in 0.010 mol/L NaF, and compare it with its solubility in pure water (2.1×10^{-4} mol/L).

7. 10.0 mL of 0.010 mol/L $BaCl_2$ is mixed with 40.0 mL of 0.0050 mol/L Na_2SO_4. Does $BaSO_4$ precipitate? Show your working.

D. Challenge

8. Milk of magnesia is a suspension of $Mg(OH)_2$. Use Le Chatelier's principle to explain why it dissolves in stomach acid (HCl).

© 2026 Chemistry Clarity · chemistryclarity.com · Free for personal and classroom use. Not for resale.